

Hamiltonian Circuit

- A Hamiltonian Circuit : is a cycle in a graph G that contains each vertex exactly once except for the starting and ending vertex that appears twice.

Examples: travelling sales man who wishes to visit every city and also minimize his route to each city and return to his home city.

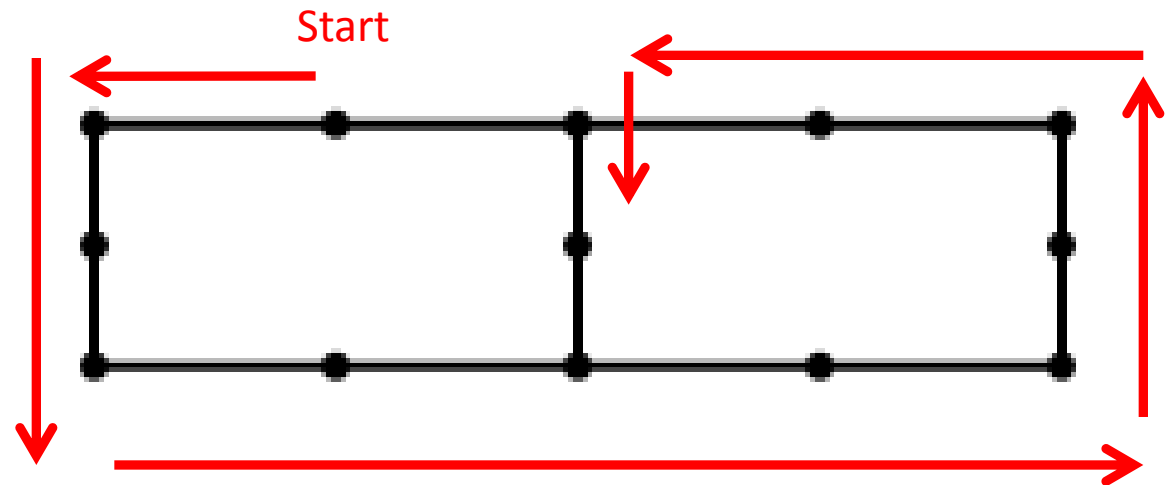
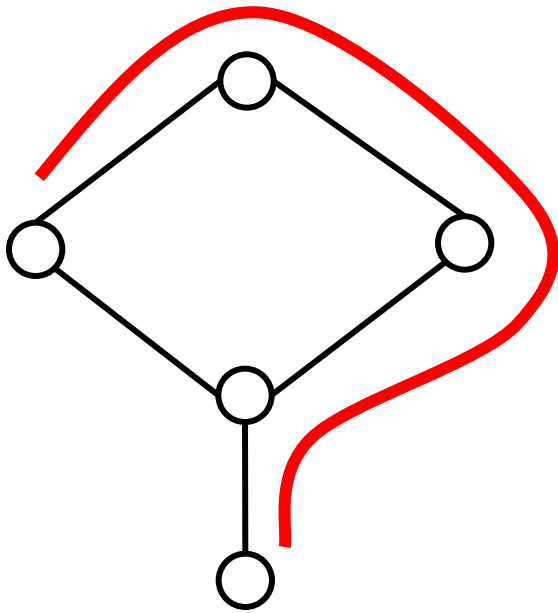
Contrast this with an Euler circuit which contains each edge exactly once.

Hamilton Path

- A Hamilton path of a graph or digraph is a path that contains each vertex exactly once.

Hamilton Path and Circuit

- Graphs with Hamilton path but no Hamilton circuit (or cycle).



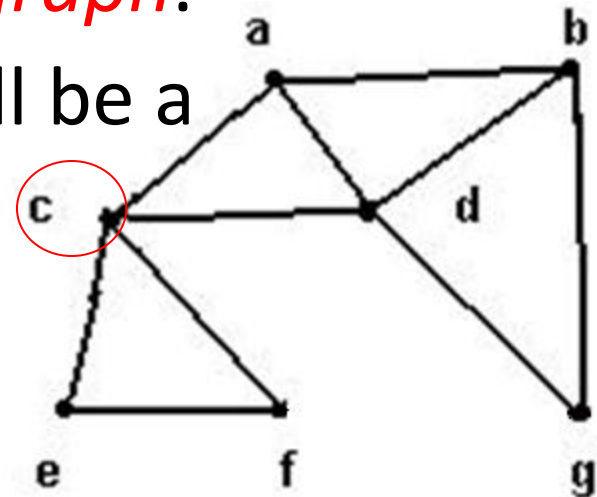
Traveling Salesman Problem

- The objective is to visit a number of cities once and return home with the minimum amount of travel.
- Used by mathematicians, statisticians, and computer scientist to solve optimization problems
- This relates to our presentation because we are going to use Hamilton circuits.

Hamiltonian cycles

Traveling salesperson problem (TSP)

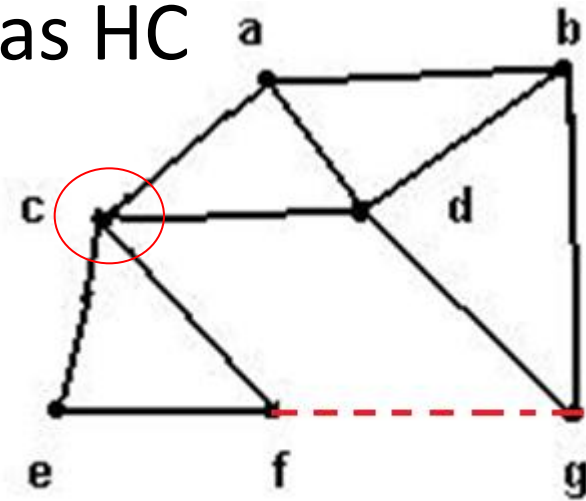
- To visit every vertex of a graph G only once by a simple cycle.
- Such a cycle is called a *Hamilton cycle*.
- If a connected graph G has a Hamilton cycle, G is called a *Hamiltonian graph*.
- A Hamilton cycle or path in G will be a *subgraph* of G .



A non-Hamiltonian graph

Hamiltonian cycles (HC)

if G has 'enough' edges, then G has HC
(add pecked red edge)



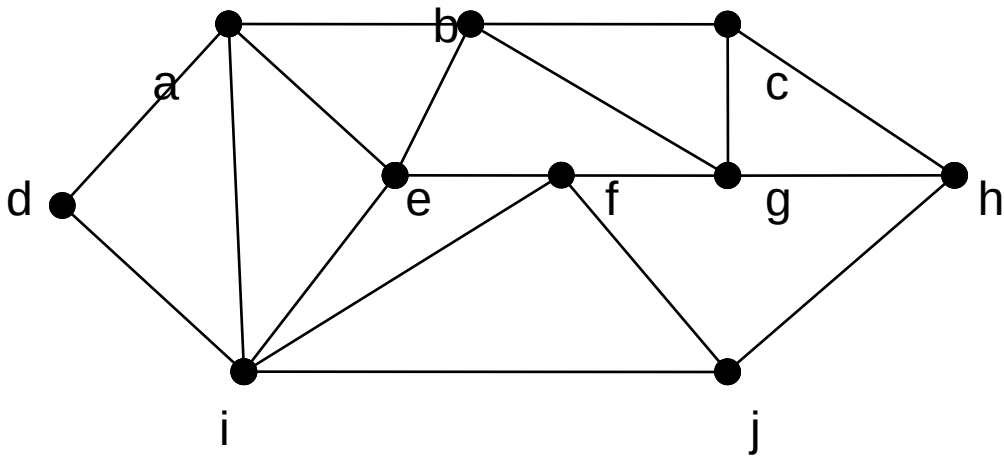
A Hamiltonian graph

Different Hamilton Circuits in Complete Graph

The number of different Hamiltonian cycles in a complete undirected graph on n vertices is $(n - 1) ! / 2$ and in a complete directed graph on n vertices is $(n - 1) !$

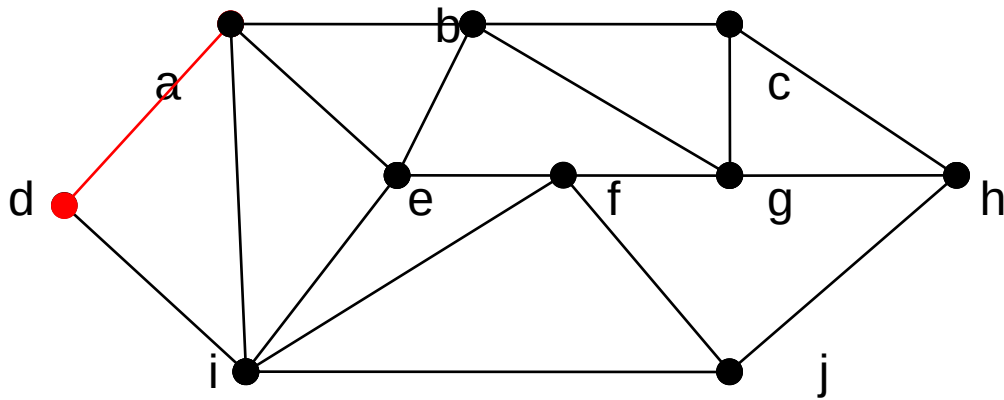
Hamilton Circuit

- Find a Hamilton circuit in the G .



Hamilton Circuit

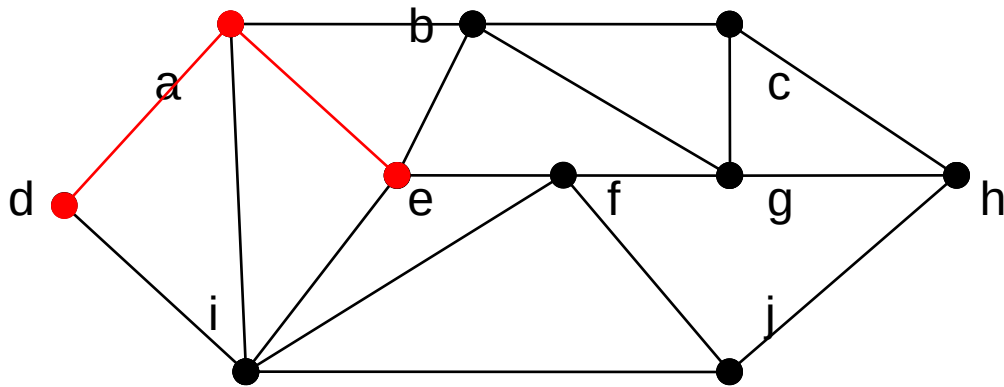
- Find a Hamilton circuit in the G.



Solution: d,a

Hamilton Circuit

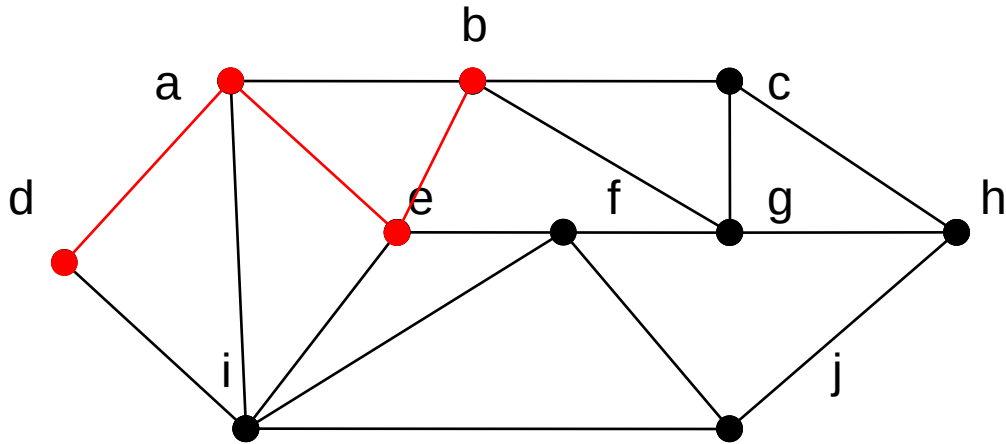
- Find a Hamilton circuit in the G.



Solution: d,a,e

Hamilton Circuit

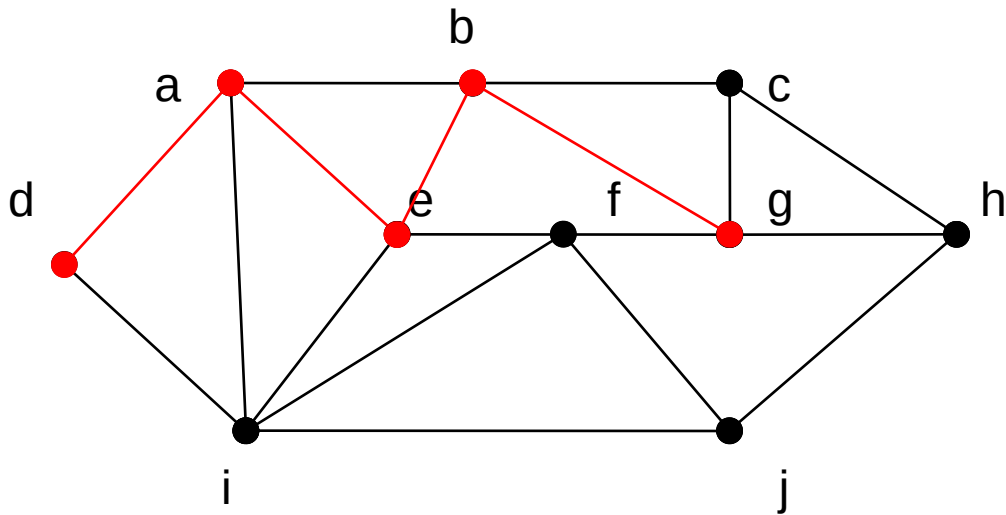
- Find a Hamilton circuit in the G.



Solution: d,a,e,b

Hamilton Circuit

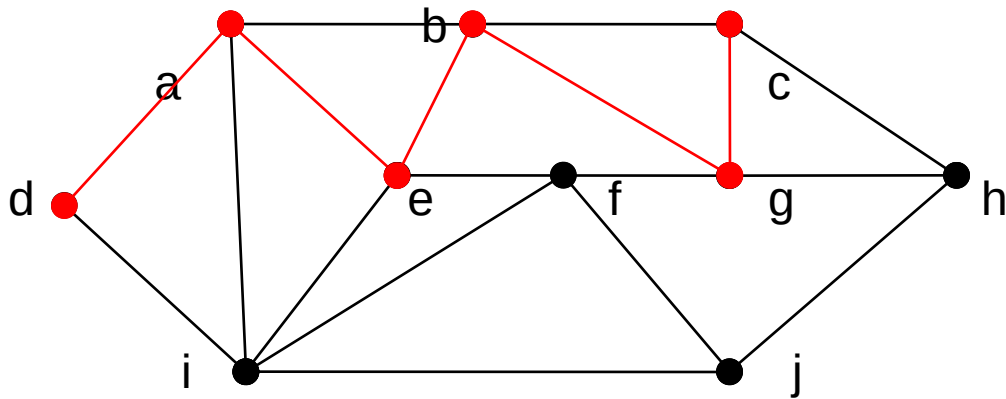
- Find a Hamilton circuit in the G.



Solution: d,a,e,b,g

Hamilton Circuit

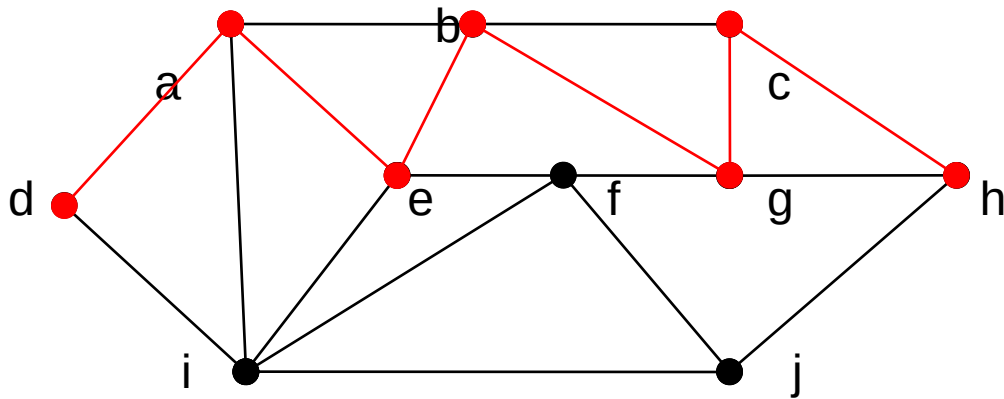
- Find a Hamilton circuit in the G.



Solution: d,a,e,b,g,c

Hamilton Circuit

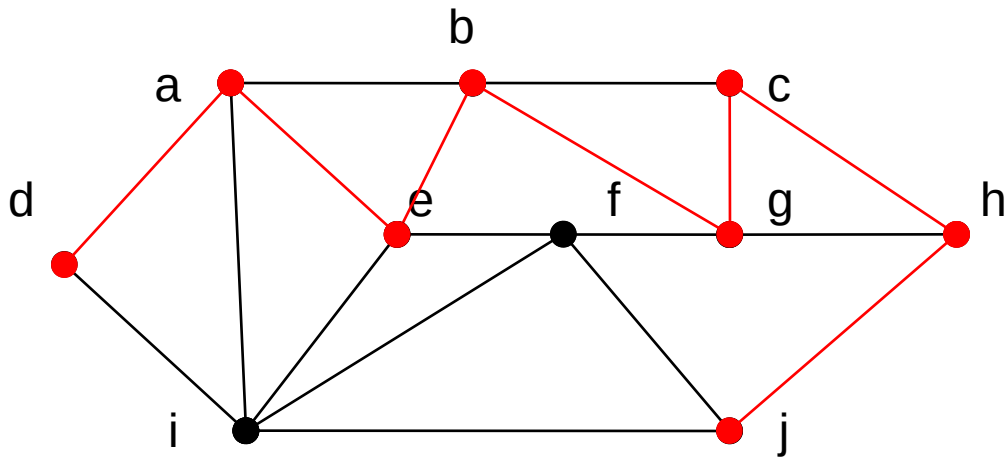
- Find a Hamilton circuit in the G.



Solution: d,a,e,b,g,c,h

Hamilton Circuit

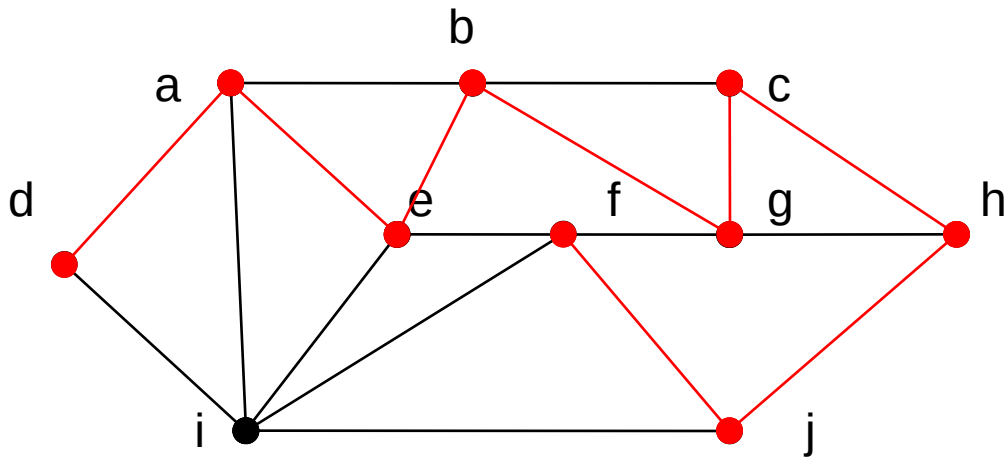
- Find a Hamilton circuit in the G.



Solution: d,a,e,b,g,c,h,j

Hamilton Circuit

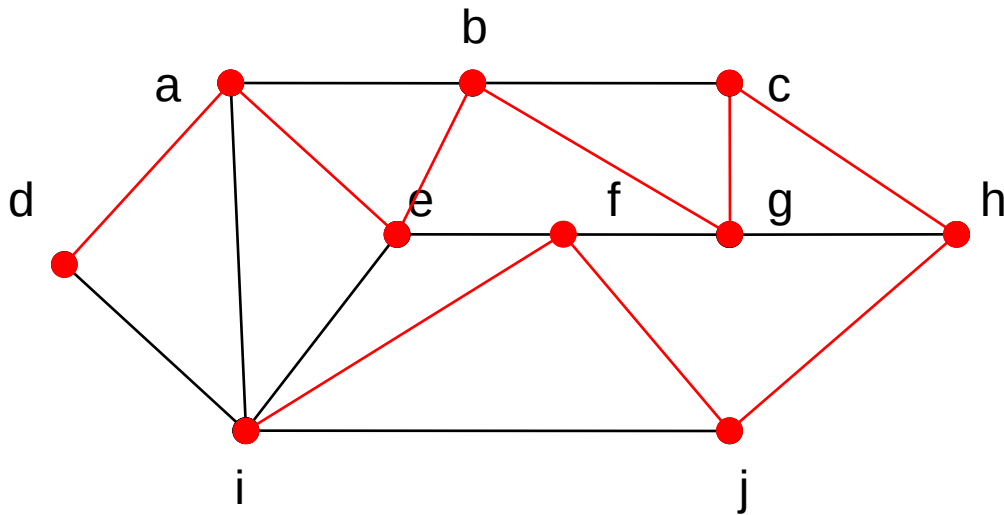
- Find a Hamilton circuit in the G.



Solution: d,a,e,b,g,c,h,j,f

Hamilton Circuit

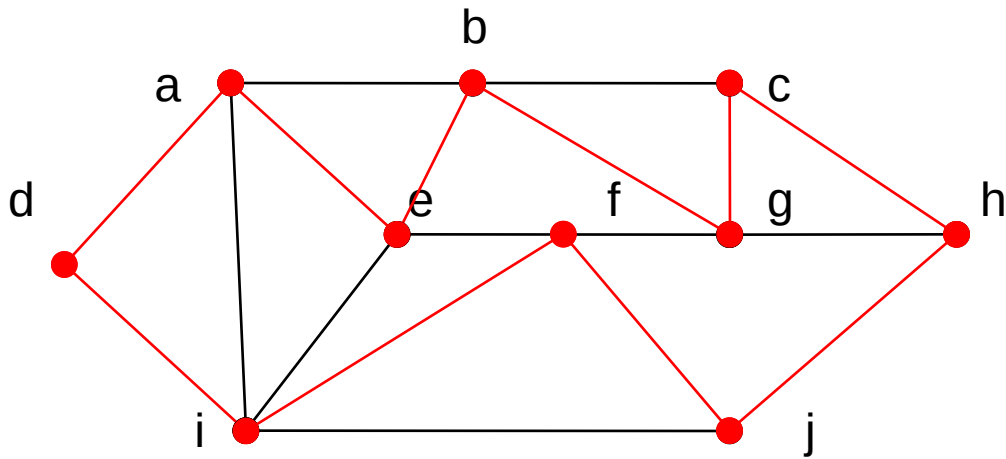
- Find a Hamilton circuit in the G.



Solution: d,a,e,b,g,c,h,j,f,i

Hamilton Circuit

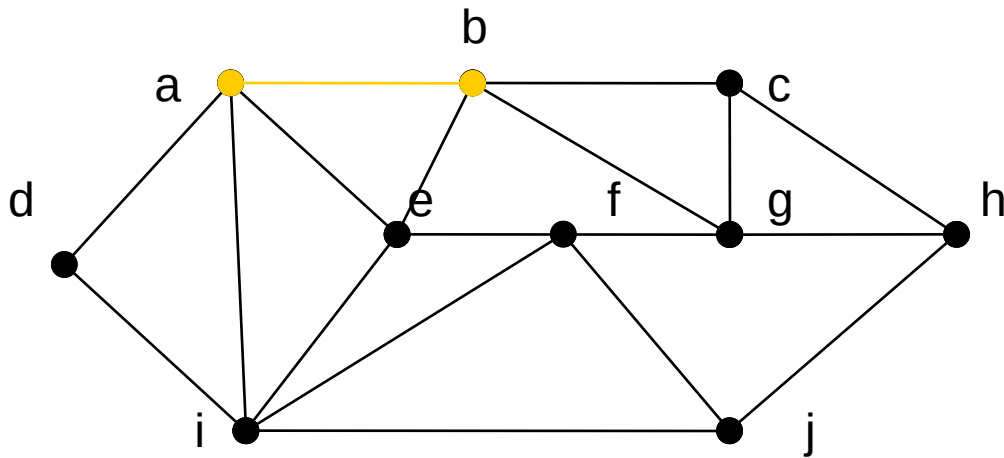
- Find a Hamilton circuit in the G.



Solution: d,a,e,b,g,c,h,j,f,i,d

Hamilton Circuit

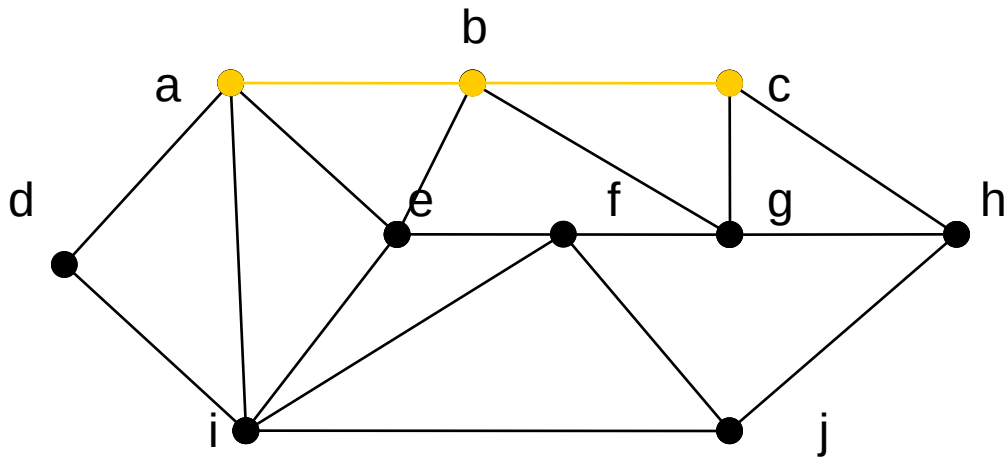
- Find another Hamilton circuit in the same graph G.



- Note that while vertices a and b were on previous Hamilton circuit the edge $\langle a, b \rangle$ was not. This is significant for TSP.

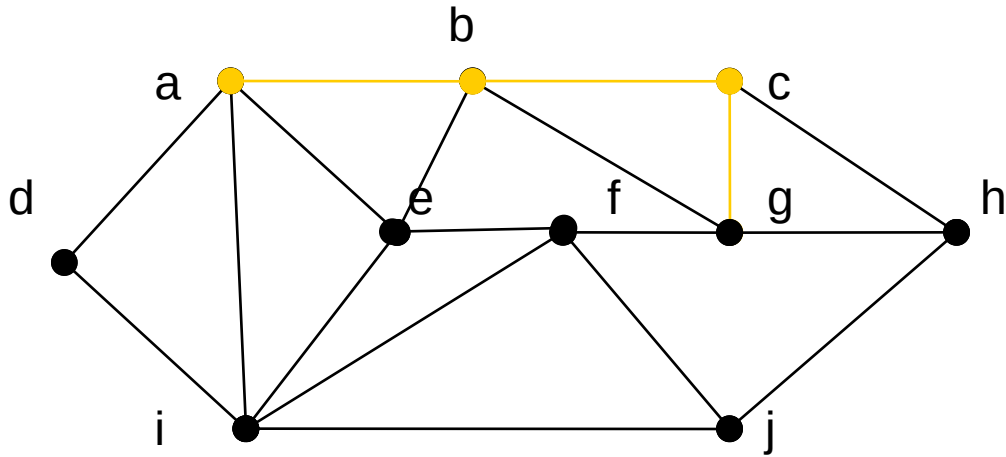
Hamilton Circuit

- Find another Hamilton circuit in the G.



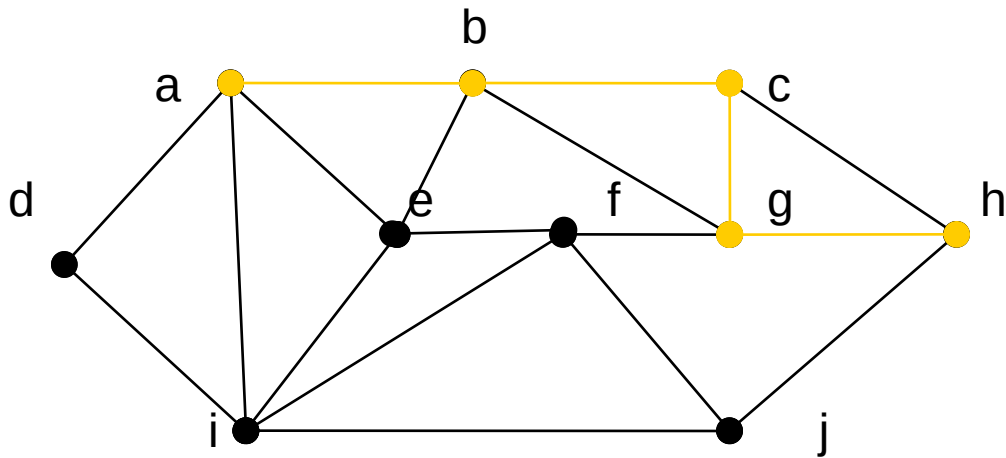
Hamilton Circuit

- Find another Hamilton circuit in the G .



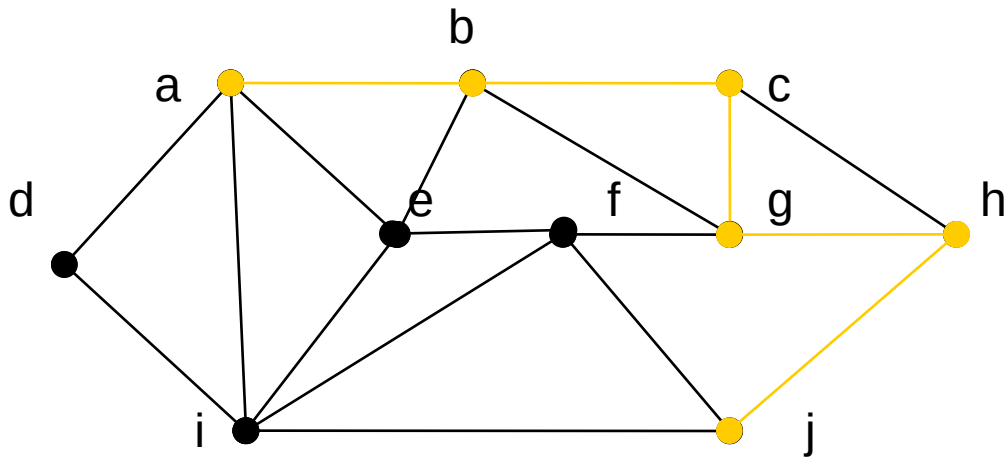
Hamilton Circuit

- Find another Hamilton circuit in the G.



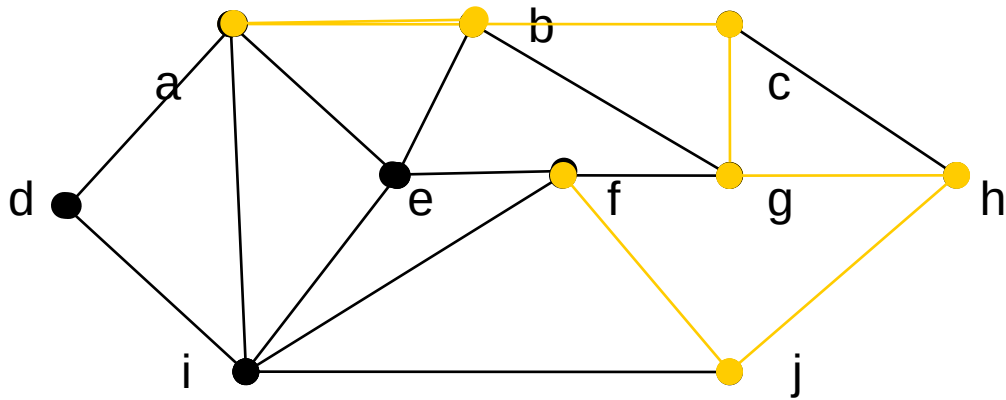
Hamilton Circuit

- Find another Hamilton circuit in the G.



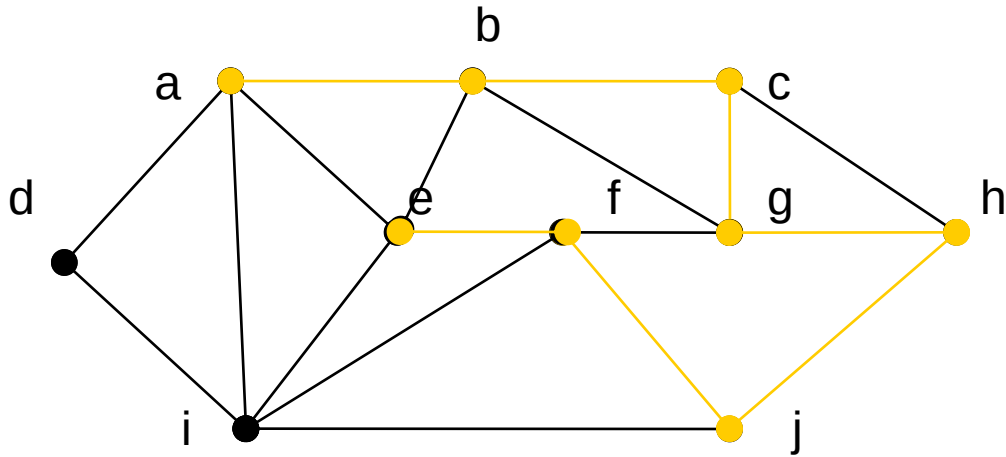
Hamilton Circuit

- Find another Hamilton circuit in the G.



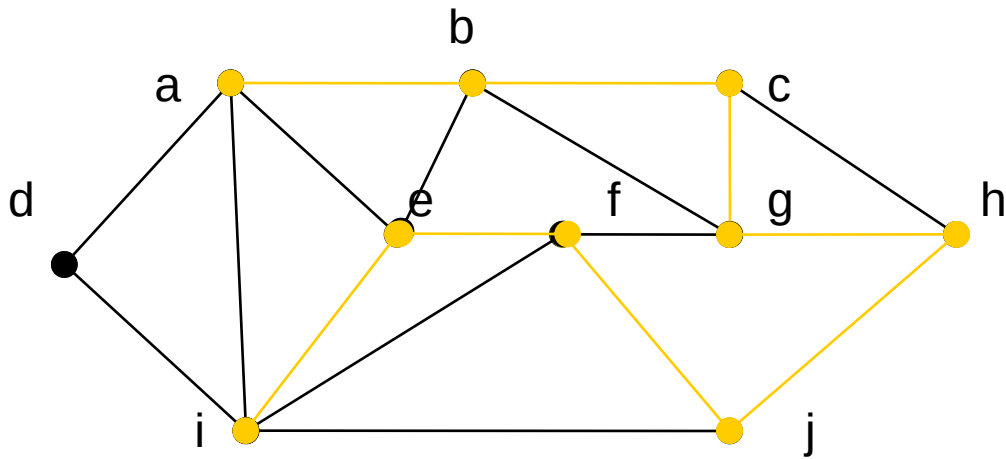
Hamilton Circuit

- Find another Hamilton circuit in the G .



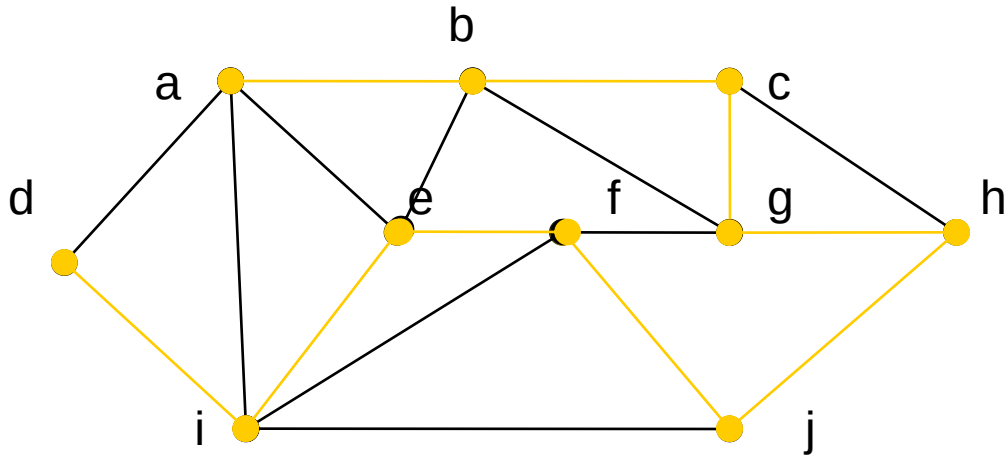
Hamilton Circuit

- Find another Hamilton circuit in the G .



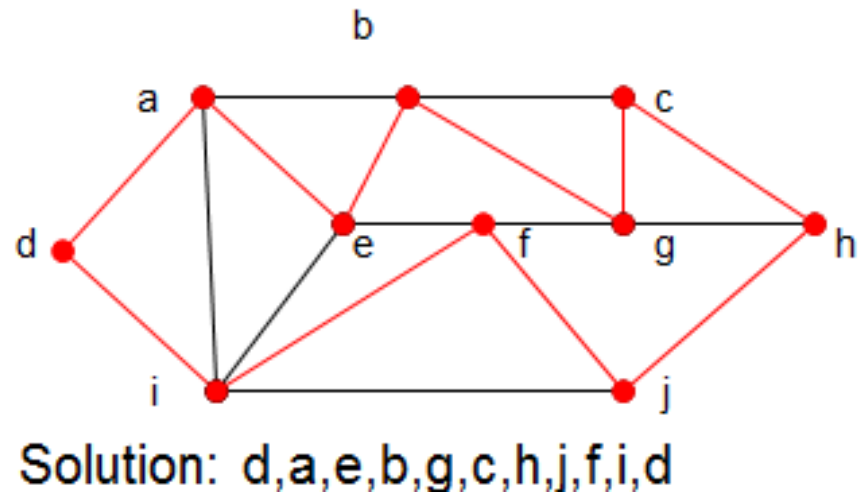
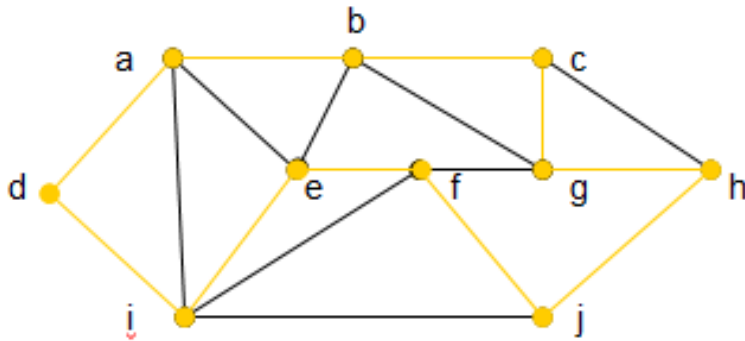
Hamilton Circuit

- Find another Hamilton circuit in the G.



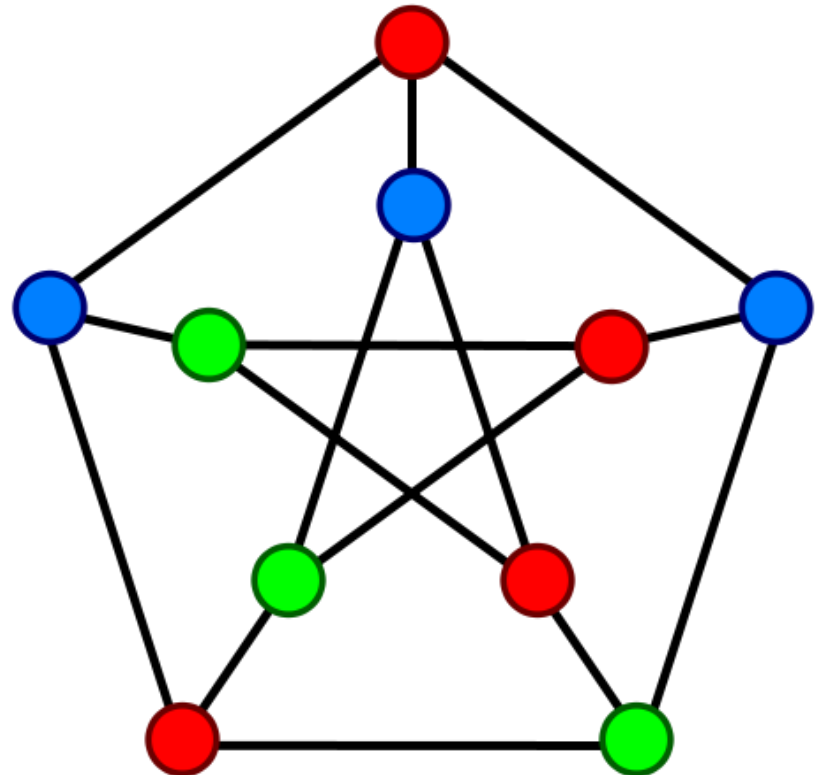
Hamilton Circuit

- Find another Hamilton circuit in the G.
- Solution $a, b, c, g, h, j, f, e, i, d, a$



Petersen Graph

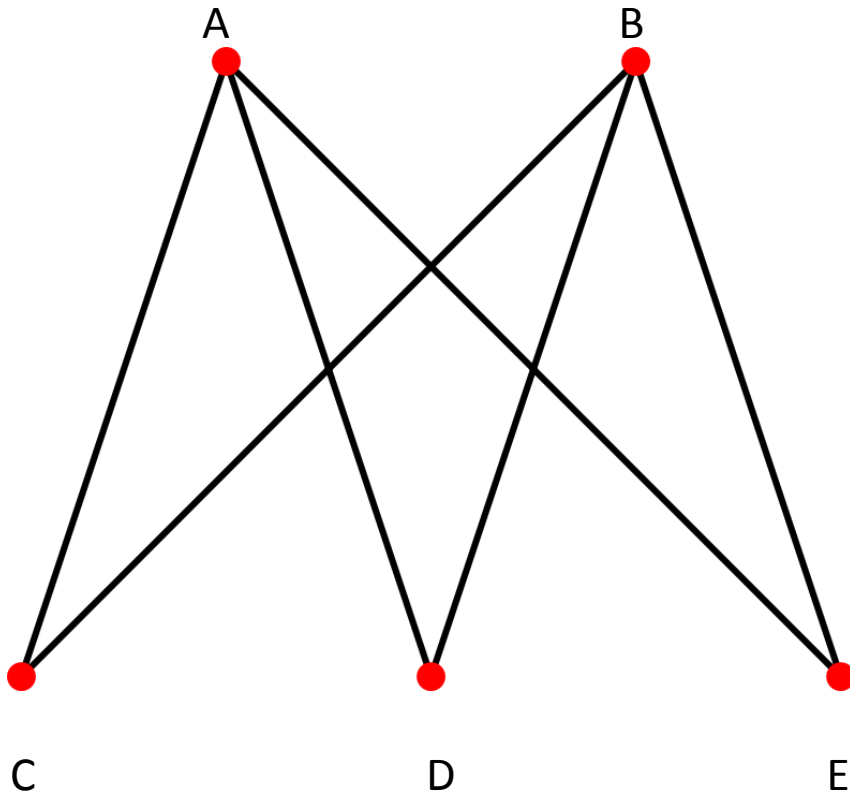
- A Petersen graph is a graph with 10 vertices and 15 edges.



DOES A PETERSON GRAPH HAVE A HAMILTON CIRCUIT?

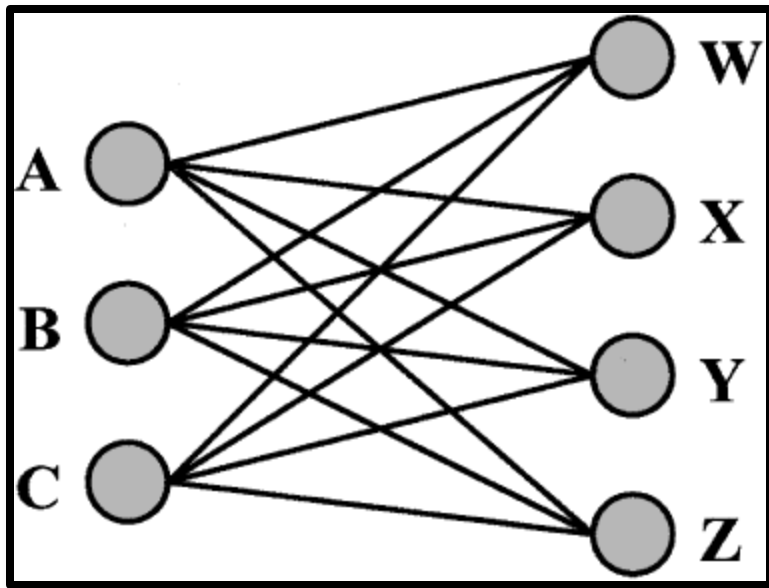
A Petersen graph has a Hamilton path but no Hamiltonian cycle.
In other words a Peterson graph does not have a Hamilton Circuit.

Bipartite Graph Ex.1



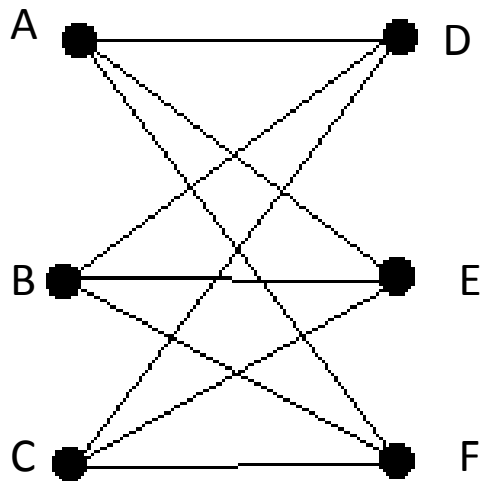
- This example has no Hamilton circuit because in a Hamilton circuit you must visit each vertex of the graph exactly once and return to the starting point. In this example you cannot do so.

Bipartite Graph Ex.2



- Also, this example has no Hamilton circuit because you must visit each vertex of the graph exactly once and return to the starting point. In this example you cannot do so.

Bipartite Graph Ex.3



- In this example, you have several Hamilton Circuits. This graph differs from the other examples because the independent sets have an equal number of points.
- Hamilton Circuits
 - A-F-C-E-B-D-A
 - D-B-F-C-E-A-D
 - B-D-A-F-C-E-B
 - C-E-A-D-B-F-C
 - E-A-D-B-F-C-E
 - F-B-D-A-E-C-F

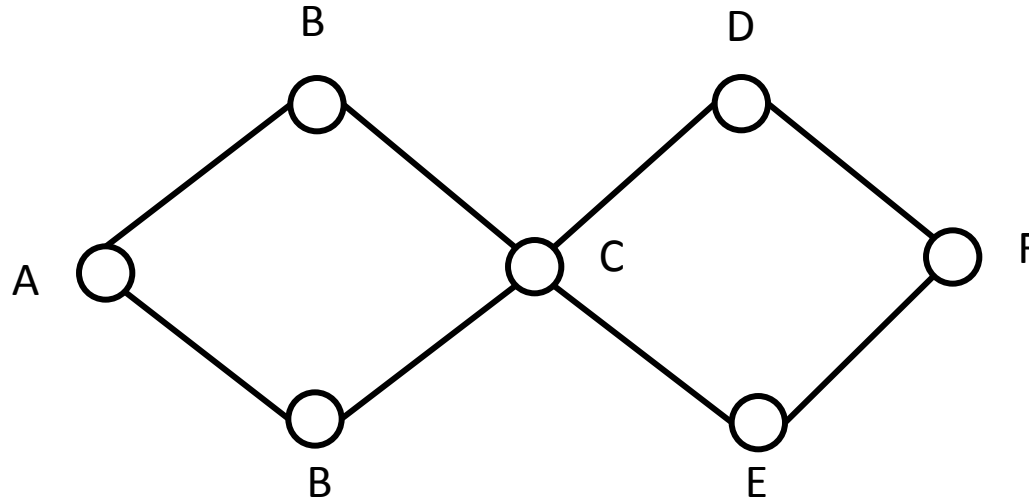
Summary of Euler and Hamilton cycles

- A **Euler cycle** is a cycle in a graph that includes all of the edges and all of the vertices in that cycle.
- A **Hamilton cycle** is a cycle in a graph G that contains each vertex exactly once except for the starting and ending vertex that appears twice.
- A path is a sequence of vertices and edges but with no edge traversed more than once
- A path with at least one edge and with the first and last vertices the same is called a cycle.

Summary of Euler and Hamilton cycles

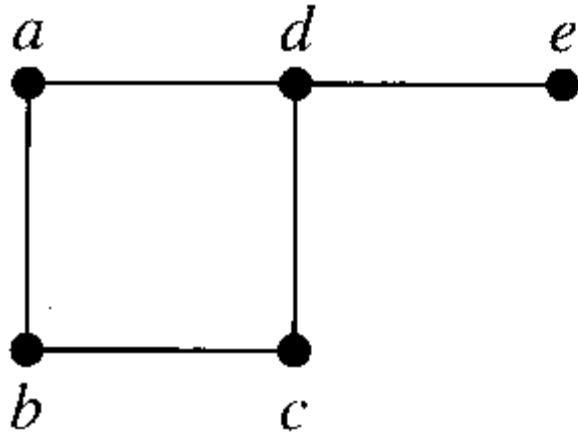
- Whether a graph does or does not have a Hamilton cycle tells you nothing about whether it has an Euler cycle , and vice versa. The same is true for Hamilton and Euler paths (rather than cycles).

Summary of Euler and Hamilton cycles

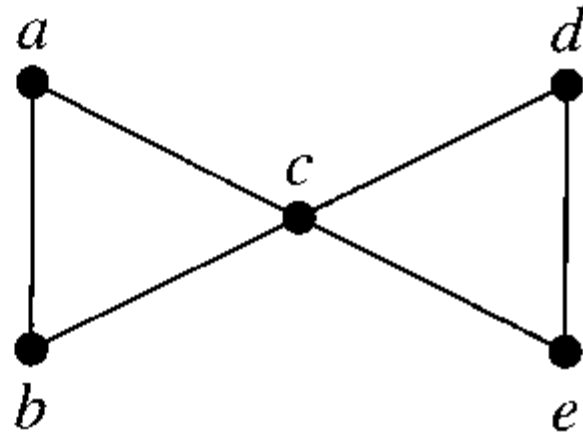


- G has a Euler cycle but no Hamilton cycle

Summary of Euler and Hamilton

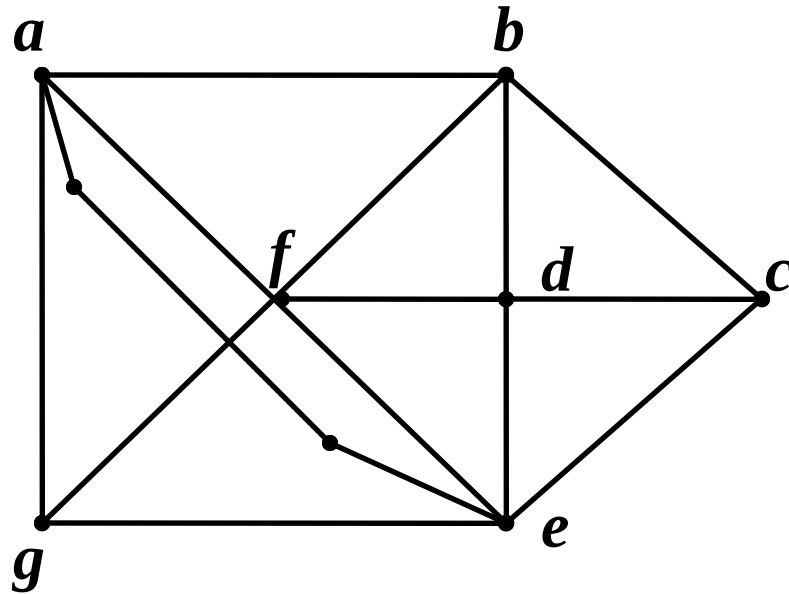


G



H

Two Graphs That Do Not Have a Hamilton Circuit.

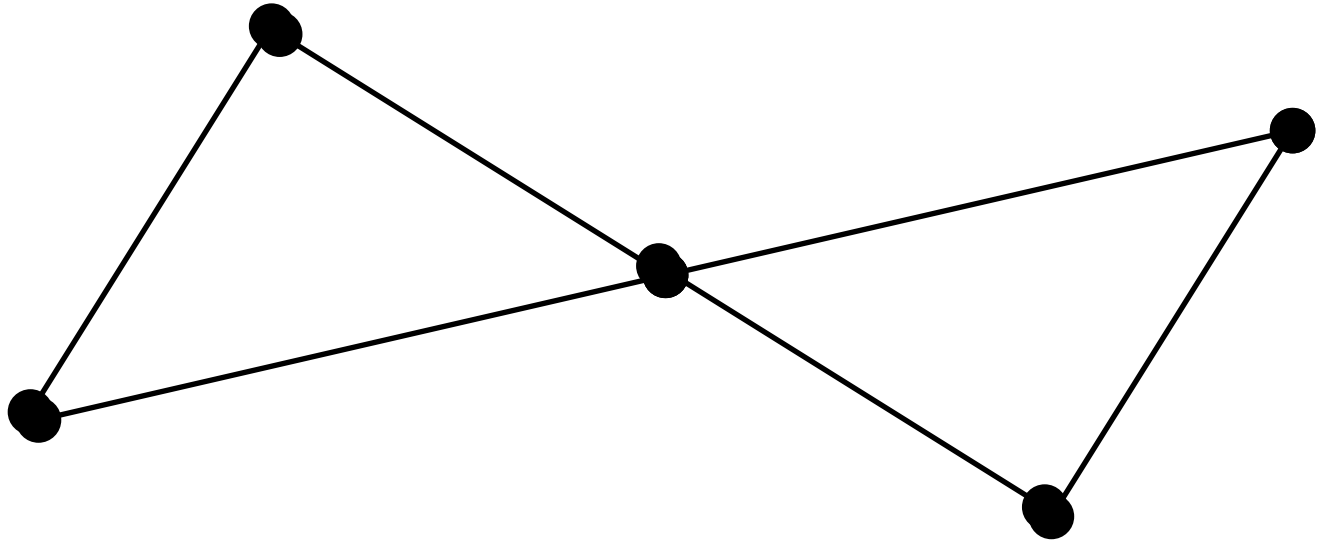


This graph G does not have a Euler cycle, since there are vertices of odd degree, yet G has a Hamiltonian cycle of:

(a,b,c,d,e,f,g,a) .

Unlike Euler cycles, no easily verified necessary and sufficient conditions are known to exist for a Hamilton cycle in a graph

edge(a,b), edge(a,f), edge(a,g), edge(a,e), edge(b,f),
 edge(b,d), edge(b,c), edge(c,d), edge(c,e), edge(e,d),
 edge(e,f), edge(e,g), edge(g,f)



This graph has a Euler cycle but no Hamiltonian cycle.

Cycle or Path?

If a graph has a Hamilton cycle does it have to have a Hamilton path?

Answer Yes: HC implies HP.

If $\langle v_1 \dots v_{n-1}, v_n \rangle$, with $v_1 = v_n$, is a Hamilton cycle then $\langle v_1 \dots v_{n-1} \rangle$, is a Hamilton path